An Introduction to Microscopes

LAB LESSON

OBJECTIVES

In this lesson, students will gain skills and knowledge regarding the use of a microscope.

ACARA CONTENT DESCRIPTIONS

Cells are the basic units of living things; they have specialised structures and functions (ACSSU149)

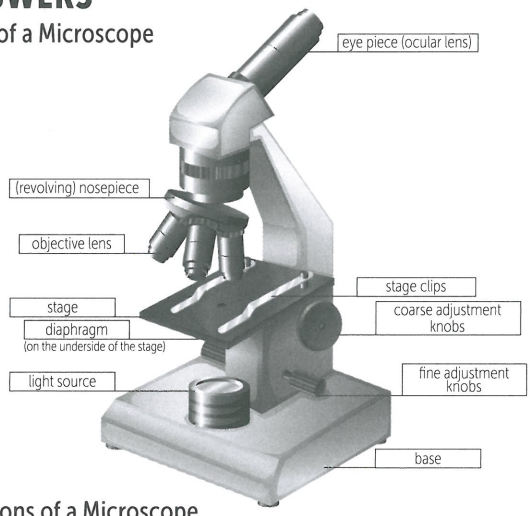
- examining a variety of cells using a light microscope

LESSON PLAN

Activities	Resources
Activity 1: What Do We Use to See? Use the following questions as discussion points: • What tool do you use to enlarge printed text that is too small for your eyes? *(Magnifying glass)* • What do people use to view things that are microscopic? *(Light microscope)* 5	
Activity 2: Parts and Functions of a Microscope Give out the Parts of a Microscope worksheet and play the video. As students watch the video, ask them to complete the worksheet. Give out the Functions of a Microscope worksheet and play the same video. As students watch the video, ask them to complete the worksheet. Use the presentation to review answers to the worksheets. 15	• Photocopies of the Parts of a Microscope and Functions of a Microscope worksheets • ClickView video *Using a Microscope - Lab Skills* **https://clickv.ie/w/lp2/2** • Presentation: An Introduction to Microscopes
Activity 3: Let's Use the Microscope Give out the Let's Use the Microscope worksheet. Play the same video from before, asking students this time to take note of the steps involved in using the microscope. Allow time for students to complete Part A of the worksheet individually. Divide students into groups of 3 and hand out the listed materials. Ask students to follow the instructions on Part B of the worksheet and see if they can find any form of life in the pond water. 40	• Photocopies of the Let's Use the Microscope worksheet • ClickView video *Using a Microscope - Lab Skills* • For each group of 3: Glass microscope slides, plastic dropper, pond water, plastic coverslip, paper towels, microscope

ANSWERS

Parts of a Microscope



Functions of a Microscope

Objective lens	Holds microscopic slide in place on the stage
Eyepiece (ocular lens)	A supportive structure that prevents the microscope from tipping over
Focus adjustment knobs (fine and coarse adjustment knobs)	Shines light up through the slide and is directed to the mirror
Stage	Collects light coming through the object and magnifies it
Stage clips	Controls the amount of light that comes through the aperture
Base	Contains the lens that allows you to view the specimen
Diaphragm	Holds objective lens in place
Light source	Moves the stage or upper part of the microscope up and down to focus on the specimen. There is a coarse adjustment knob (used first) and a fine adjustment knob (for fine-tuning the focus)
(Revolving) nosepiece	Holds and supports microscope slides containing the specimen

Let's Use the Microscope

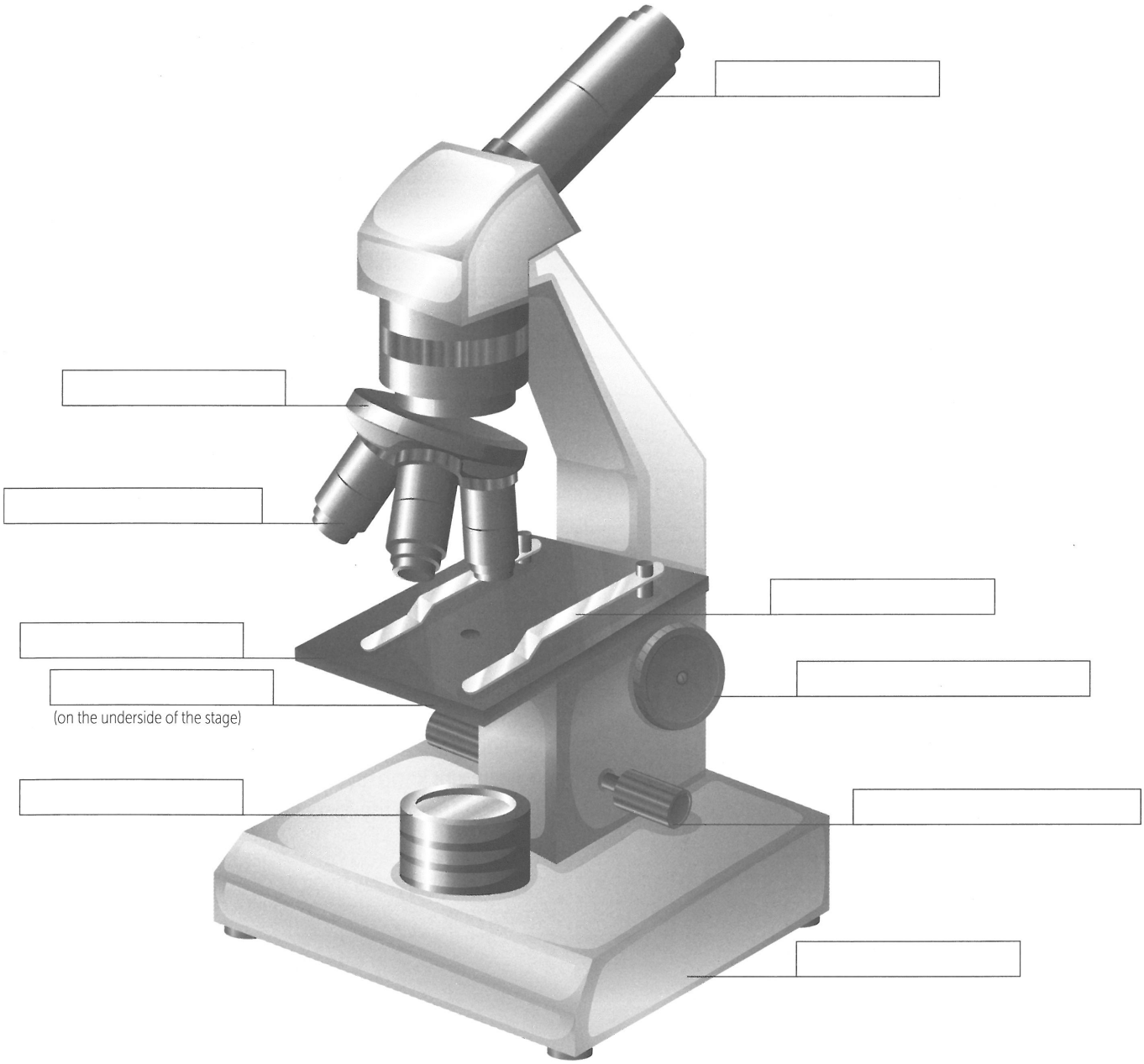
1	Carry the microscope with two hands with one hand on the base.
2	Always begin focusing with the lowest power objective lens (e.g. 4x) and the stage in the lowest position so that the slide never touches the objective lens (this avoids cracking of the glass slides and coverslips, and damaging the lens).
3	Use the coarse adjustment knob to move the stage up and focus the specimen.
4	Use the fine adjustment knob to adjust visibility of the specimen, or change to a higher objective lens.
5	View the specimen under the microscope.
6	Lower the stage and then remove the slide when finished.
7	Return the lowest power objective lens into position over the stage, and be sure the stage is at its lowest function when you have finished using the microscope.
8	Turn off the light and wrap the cord correctly before putting it away.

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Parts of a Microscope

Label the parts of the microscope with the words in the box below using information from the video and the presentation.

base	fine adjustment knobs	stage	stage clips	eyepiece (ocular lens)
diaphragm	objective lens	coarse adjustment knobs	light source	(revolving) nosepiece

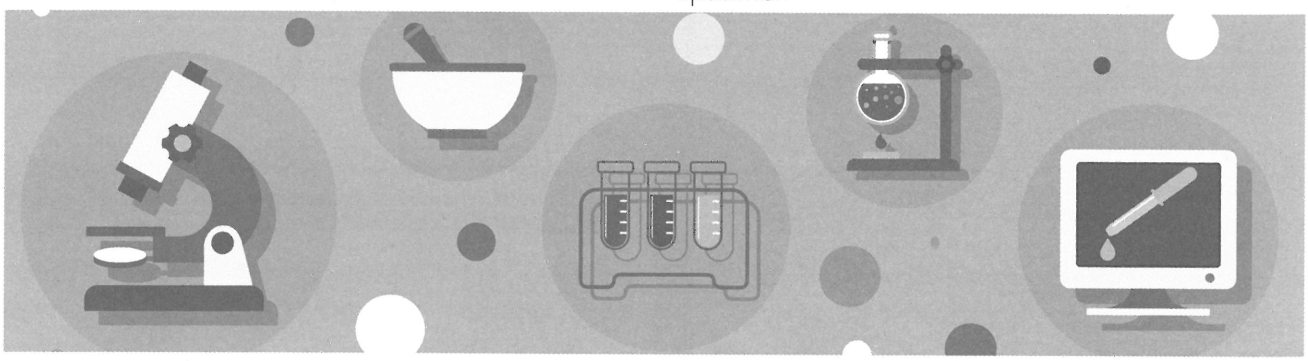


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Functions of a Microscope

Draw a line to match the features of a microscope to their function. Use the information in the video and the presentation to assist you.

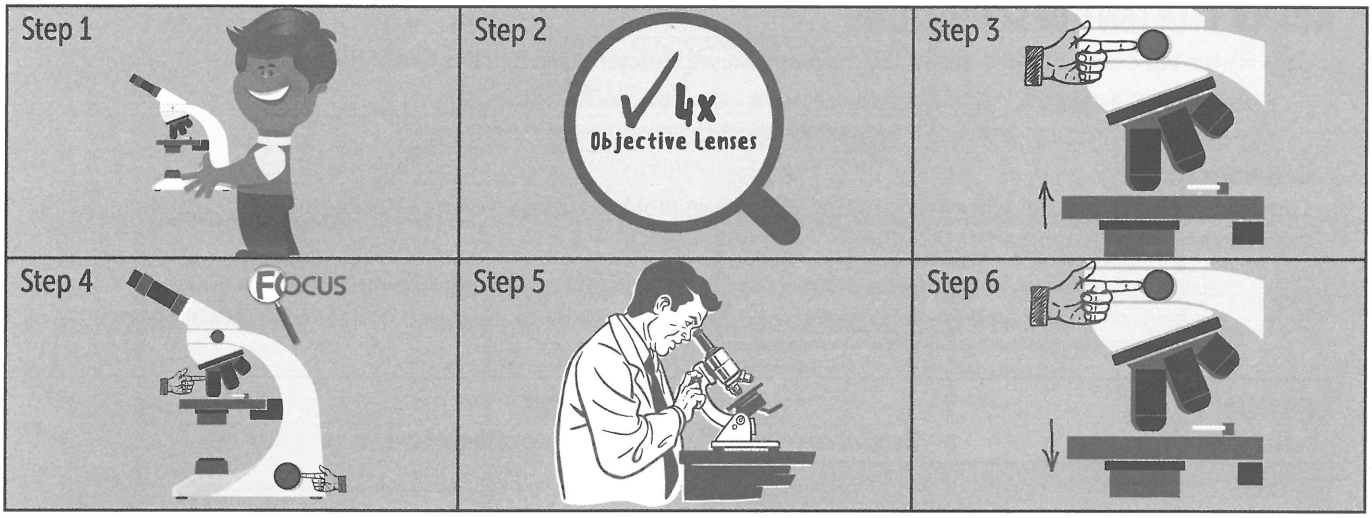
Objective lens	Holds microscopic slide in place on the stage
Eyepiece (ocular lens)	A supportive structure that prevents the microscope from tipping over
Focus adjustment knobs (fine and coarse adjustment knobs)	Sends light up to the slide and is directed to the mirror
Stage	Collects light coming through the object and magnifies it
Stage clips	Controls the amount of light that comes through the aperture
Base	Contains the lens that allows you to view the specimen
Diaphragm	Holds objective lens in place
Light source	Moves the stage or upper part of the microscope up and down to focus on the specimen. There is a coarse adjustment knob (used first) and a fine adjustment knob (for fine-tuning the focus)
(Revolving) nosepiece	Holds and supports microscope slides containing the specimen



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Let's Use the Microscope

Part A: Using the pictures as clues, create an instructional manual on how to use a microscope. Some of the steps have been provided to assist you.



Step 1	
Step 2	
Step 3	Use the coarse adjustment knob to move the stage up and focus the specimen.
Step 4	
Step 5	View the specimen under the microscope.
Step 6	
Step 7	Return the lowest power objective lens into position over the stage, and be sure the stage is at its lowest function when you have finished using the microscope.
Step 8	Turn off the light and wrap the cord correctly before putting it away.

Part B: Prepare a slide and observe it under the microscope. Answer the question in the box below.

Materials:

- microscope
- glass microscope slide
- plastic dropper
- pond water
- plastic coverslip
- paper towels

Instructions:

1. Place a drop of pond water onto the microscope slide using a pipette or dropper.
   TIP: Too small a drop may result in the specimen in the water being crushed. Use the paper towel to remove excess water from the slide if necessary.
2. Place a coverslip on top of the specimen and observe it under the microscope.

What did you see under the microscope?

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